

LAB-03 Component Diagram

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Class: AIML-D

Date: 08/10/2026

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Alumni Mentorship & Mock Interview Platform

Architecture Selection: We chose Client-Server Architecture for the Alumni Mentorship & Mock Interview Platform.

Why not Layered: Layered architecture organizes a single application into internal tiers (presentation/business/data) — it doesn't naturally model two distinct client apps (Student Mentee, Alumni Mentor) talking to one shared backend over the network, and it says nothing about how both roles must share the same live availability calendar, which is exactly the access pattern this system needs.

Why not Microservices: Splitting Mentor Matching, Session Booking, and Interview Scorecards into independently deployed microservices adds network hops and operational complexity that a campus-scale platform doesn't need. Worse, guaranteeing that a booked slot is no longer available to another student (FR-003) would then require distributed locking or sagas across services — a Client-Server model gives that consistency for free with a single atomic transaction on one server.

Reason 1 — Single source of truth for mentor availability (no double-booking)

The acceptance criterion of FR-003 is that once a slot is booked it is no longer available to another student, even when many students try to grab the same popular mentor's slot at the same moment (e.g., during placement season) — exactly the case handled by alternate flow 6a of UC-03. In a Client-Server model, the Session Booking & Calendar Server owns the one authoritative slot table and confirms each booking inside a single atomic transaction (row-level lock on the slot). Exactly one request wins; every other client is told the slot is taken and shown the mentor's remaining slots. Both the Student Mentee Client App and the Alumni Mentor Portal Client always see the same, consistent calendar (FR-002), and both receive the same booking confirmation.

Reason 2 — Centralized control over private profiles and interview scorecards

Mock-interview scorecards are sensitive evaluative data about students, and profiles contain personal and employer details of alumni. A Client-Server model keeps all profile, session, and scorecard records and the business rules on central servers, so the NFR-001 rule that authenticated users can access only the information they are authorized to view (e.g., a scorecard is visible only to the student and the mentor who conducted the interview, as required by FR-004) is enforced in one place, together with audit logging (mentor ID + timestamp on every scorecard submission), rather than duplicated or drifting across independently deployed services.

Security Advantage

Because the Student Mentee Client App and the Alumni Mentor Portal Client are thin clients that hold no persistent copies of profiles or scorecards, a lost or compromised device does not expose the underlying database. Every request to the Profile & Mentor Matching Server and the Interview Scorecard Server passes through TLS 1.3 in transit and is checked against the user's session token and role-based access policy before any data is returned; data at rest is encrypted with AES-256. Privacy requests such as soft-deletion of a profile or scorecard are carried out once, on the server, concentrating the system's security and privacy boundary at the server rather than across many client endpoints.

Performance Benefit

Centralizing recommendation on the Profile & Mentor Matching Server lets it maintain a shared, precomputed index of mentors by domain, skills, and areas of interest (Embedded Systems, Cloud Computing, Data Science, etc.) and a cached view of their free slots (refreshed through IMentorAvailability) in server memory, instead of rebuilding it per client request. This server-side optimization (in-memory domain index, caching, connection pooling) is what keeps the core operations named in NFR-001 — mentor search and session booking — responsive under simulated peak load, since the expensive matching work is done once centrally and a booking is a single local database transaction with no cross-service calls. Fast, consistent responses from one server also support NFR-002, as calendar availability and booking status shown to users are always current.

Requirements Traceability — FR/NFR to Architecture Mapping

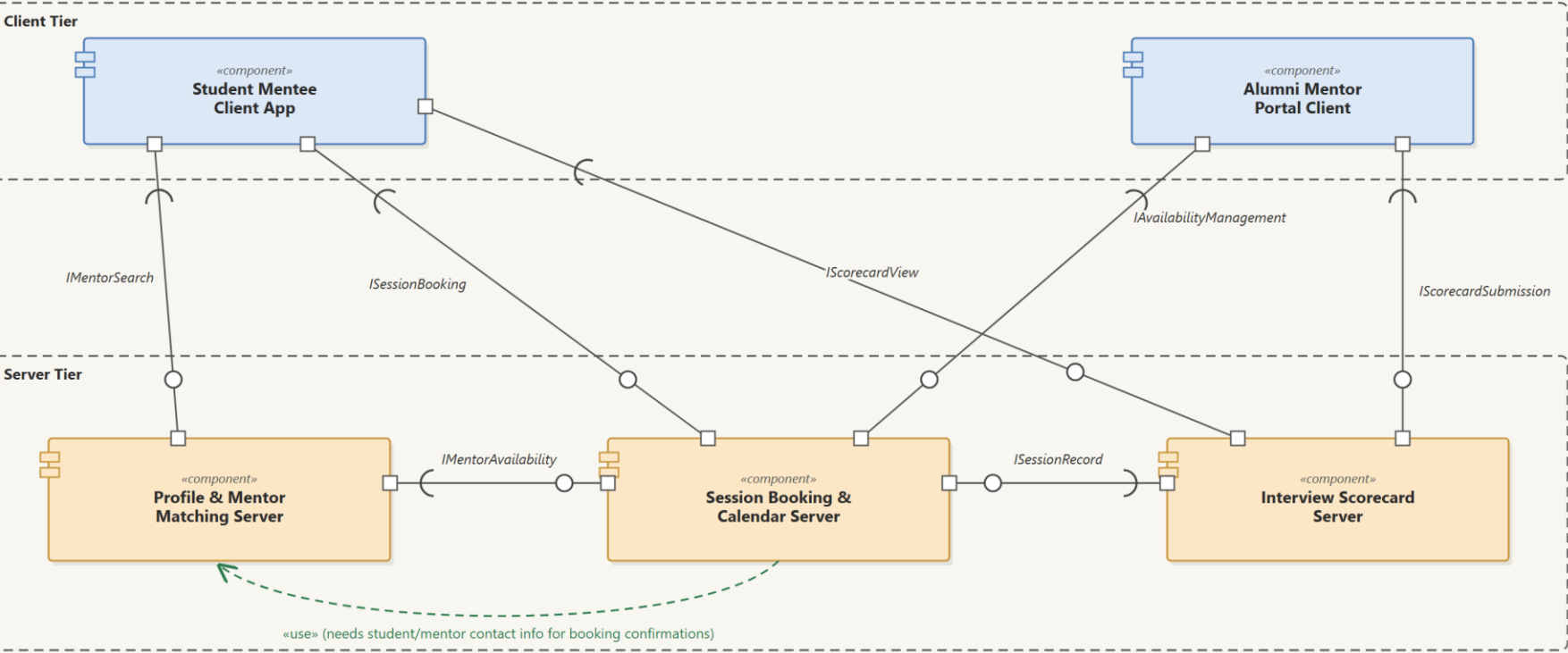
ID	Type	Description	Priority	How the Architecture Addresses It
FR-001	Functional	Recommend alumni mentors to students based on the selected domain, skills, areas of interest, and mentorship requirements, and let the student select a suitable mentor.	High	Student Mentee Client App calls IMentorSearch on the Profile & Mentor Matching Server, which ranks mentors whose listed domain matches the student's and filters out fully-booked mentors using IMentorAvailability from the Session Booking & Calendar Server.
FR-002	Functional	Alumni mentors create, view, and update their available dates and time slots for mentorship and mock interview sessions.	High	Alumni Mentor Portal Client invokes IAvailabilityManagement on the Session Booking & Calendar Server — the single authoritative calendar, so an updated slot is immediately bookable by eligible students.
FR-003	Functional	Student books a mentorship or mock interview session by selecting an available mentor and calendar slot; the booked slot is no longer available to others.	High	Student Mentee Client App calls ISessionBooking; the Session Booking & Calendar Server locks the slot in one atomic transaction, creates the booking, and uses the «use» dependency on the Profile & Mentor Matching Server to fetch contact details for confirmations.
FR-004	Functional	Alumni mentors record a structured scorecard (ratings, comments, feedback) after a student's mock interview; it is accessible to the associated student.	High	Alumni Mentor Portal Client submits via IScorecardSubmission to the Interview Scorecard Server, which checks the session against ISessionRecord from the Booking Server and saves the scorecard against it.
FR-005	Functional	Students view their confirmed sessions, mock interview history, and completed post-interview scorecards.	Medium	Student Mentee Client App reads upcoming and past sessions via ISessionBooking and the attached scorecards via IScorecardView, so the dashboard is assembled from the two authoritative servers.

ID	Type	Description	Priority	How the Architecture Addresses It
NFR-001	Performance & Security	Protect profiles, session information, and post-interview feedback per privacy guidelines while staying responsive under simulated peak load.	High	Access control, TLS 1.3, and AES-256 are enforced server-side, and clients hold no persistent copies; central caching and connection pooling keep mentor search and booking responsive at peak load.
NFR-002	Usability & Availability	Provide a straightforward, responsive interface; calendar availability and booking status are clearly displayed.	High	Both thin clients render from one authoritative calendar on the Session Booking & Calendar Server, so the availability and booking status users see is always current and consistent.

Component Diagram

Alumni Mentorship & Mock Interview Platform

UML Component Diagram — Client-Server Architecture



Legend

Provided Interface

Required Interface

Port

«use» Dependency

5 components • 7 interfaces (*IMentorSearch*, *ISessionBooking*, *IScorecardView*, *IAvailabilityManagement*, *IScorecardSubmission*, *IMentorAvailability*, *ISessionRecord*) + 1 «use» dependency